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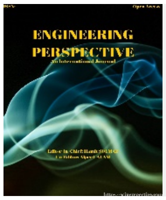


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Research Paper

# Detection of Retinal Dysfunction with Multimodal PERG Analysis: A Patient-Level Hybrid Machine Learning Framework

Yavuz Bahadır Koca[1]*

[1] *Electrical Engineering Department, Faculty of Engineering, Afyon Kocatepe University, Afyonkarahisar, 03200, Türkiye*

## ABSTRACT

Pattern electroretinogram (PERG) is the standard for assessing retinal ganglion cell function. However, the low amplitude and complex waveform of PERG signals complicate clinical interpretation. This study proposes a robust, multimodal hybrid machine learning framework that detects retinal dysfunction under a rigorous patient level validation strategy by integrating PERG waveform features with clinical demographic data. The PERG-IOBA dataset, consisting of 1354 signals from 304 participants was used. Training and test sets were separated at the patient level using 5-fold cross validation to approximate real clinical deployment and to avoid information leakage. A dual-stream model was developed. One stream processed functional PERG features, latency, amplitude and RMS via a multilayer perceptron, while the second stream processed clinical data. The two representations were then fused at the feature concatenation level. This model (Model 1) was compared with a stacking ensemble of conventional classifiers (Model 2) and a two-stage cascade classifier tailored for screening (Model 3). Model 2 achieved the most balanced and robust performance with 71.4% accuracy and an Area Under the Curve of 0.76 in 5-fold patient level cross validation. Although more modest than many previously reported values, these metrics are consistent with realistic clinical generalizability. The Model 3 provided the highest sensitivity with 79.7% for screening purposes. SHAP analysis confirmed P50-N95 amplitude as the primary biomarker but identified age as a significant confounding factor, mimicking expert clinical judgment. This study demonstrates that retinal dysfunction detection requires a whole approach that integrates signal morphology and patient demographics.

**Keywords:** Machine Learning; Pattern Electroretinogram; SHAP Analysis; XAI; Feature Fusion; Stacking Ensemble

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**Author Contacts :** ybkoca@aku.edu.tr, ybkoca@gmail.com

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*Corresponding Author

## 1. Introduction

Electroretinogram (ERG) is a non-invasive diagnostic method that measures the electrical response of retinal cells to light stimuli. The ERG test evaluates potentials generated directly by retinal neurons and mediated by glial cells. Thus, it provides information about retinal function [1]. In clinical practice, thin fiber or contact lens type electrodes placed on the corneal surface are typically used. These electrodes record the electrical activity generated by the retina. ERGs can be obtained using various methods, including full field flashes, multifocal stimuli or patterned stimuli. Recording protocols have been set up by the International Society for Clinical Electrophysiology of Vision (ISCEV). Standards guarantee the reproducibility and reliability of measurements [2,3].

ERG testing is a physiological signal monitoring (PSM) method used in the diagnosis and monitoring of eye diseases. The ERG test is useful for diagnosing retinitis pigmentosa, cone rod dystrophies, toxic retinopathies, and diseases of the optic nerve. It may also be utilized in instances of drug-induced retinal toxicity or foreign body penetration into the eye [4,5]. There are different ways to measure ERG depending on the type of stimulus and how it is recorded. These include focal ERG (fERG), full field ERG (ffERG), multifocal ERG (mfERG) and pattern ERG (PERG). ffERG is used to analyze the functional integrity of the entire retina [6].

mfERG is used to detect local dysfunction in the macula [7]. fERG is also a method for assessing functional disorders in the central visual field [8]. PERG is specifically used to evaluate the function of macula and retinal ganglion cells [9]. PERG is an important diagnostic method in eye diseases. It is used in the diagnosis and monitoring of various eye diseases. PERG provides information for a more accurate diagnosis of central retinal function. It is important to distinguish macular and optic nerve disorders. It is also a noninvasive method for assessing retinal health and function. In this context, the clinical significance of PERG lies in its ability to detect functional changes in the retina before structural changes occur. Thus, PERG testing allows for early diagnosis and treatment of retinal diseases. PERG is generated with a contrast reversing pattern, usually a black and white checkerboard pattern [10,11]. An application of the PERG is given with Figure 1.

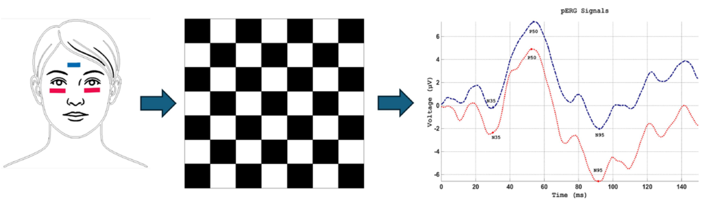

Figure 1. PERG application

Significant challenges arise in analyzing PERG signals in the clinic. PERG signals, by their nature, typically have low amplitudes, typically in the microvolt range [12]. In addition, these signals can be easily affected by disturbances such as eye blinking, muscle movements, or electrical noise. This can lead to differing perspectives among experts when interpreting the signals. Therefore, manual analysis makes the detailed evaluation of PERG signals time consuming. In recent years, notable advances have been made in PSM methods. One of the most important parameters in this area is the integration of artificial intelligence (AI) and wearable sensor technologies. In this context, the challenges in calculating and analyzing PSMs can be addressed more successfully and easily [9,13,14]. Especially, machine learning (ML) approaches have the potential to increase diagnostic accuracy in retinal function assessment [15,16]. Diseases such as glaucoma and diabetic retinopathy are experiencing a significant increase in cases worldwide [17]. Hereby, AI based automated analysis systems can play a significant role in improving the quality of healthcare and reducing workload [18]. So, AI is being used extensively in many fields recently, from education and healthcare to energy [19].

Two critical limitations can be noted in the existing literature on AI-based PERG analysis. First, most existing studies rely solely on signal morphology and ignore the entire clinical context. Most existing studies typically use hand crafted parameters such as N35, P50 and N95 for classification. While this analysis reflects physician knowledge, it can lead to the loss of morphological details and temporal patterns hidden throughout the signal waveform. In the clinic, ophthalmologists do not evaluate PERG waveforms in isolation [12–15]. In accordance with ISCEV standards, they interpret these signals within the context of the patient's demographic profile and visual acuity [3,11]. AI models based solely on signals lack this holistic perspective. Consequently, they can lead to misinterpretations in borderline cases where age-related signal deterioration can be confused with disease [16,17]. Second, many DL studies report nearly perfect classification accuracies, often above 95%, which are likely inflated due to inappropriate validation schemes [18,20]. Studies that combine the characteristics of both the raw signal and expert parameters into a method that dynamically determines the characteristics are quite limited.

The main objective of this study is to use waveform parameters derived from pattern electroretinography signals of the PERG-IOBA dataset and eye-level clinical variables to classify normal and pathological eyes. For this purpose, PERG waveform features (latency, amplitude differences, ratios, and time-domain summaries) are integrated with clinical indicators such as age, sex, and logMAR visual acuity. Three different hybrid ML architectures are evaluated: (i) a dual-stream network that processes PERG and clinical inputs in two separate streams and combines them into a common representation space; (ii) a stacked ensemble structure that combines the outputs of random forest (RF), XGBoost, LightGBM and multilayer perceptron (MLP) with a logistic regression (LR) meta-model; and (iii) a two-stage cascade classifier that rapidly extracts high-confidence normal/pathological cases and directs cases in the gray zone to a more complex hybrid model. In this study, we systematically examined both the generalization success of these architectures and the diagnostic contribution of PERG and clinical features using patient level group cross validation and SHapley Additive Explanations (SHAP) based explainability analysis.

This study contributes to the literature in three key ways. First, a patient level cross validation scheme was used to prevent data leakage unlike standard ML algorithms. This provided a realistic estimate of clinical performance rather than empirical metrics. Second, the model captures multidimensional interactions by combining signal parameters with clinical data. This mimics expert decision-making processes. And third, the influence of features was quantitatively demonstrated with using SHAP. The clinical interpretability of black box models was enhanced by identifying age as a specific confounding factor.

## 2. Materials and Methods

A feature-based ML methodology was developed for normal and diseased eye analysis using PERG recordings from the PERG-IOBA dataset. The methodological process consisted of four main structures. First, a PERG database with eye-level clinical results was used. In the second stage, raw PERG signals were preprocessed and N35, P50 and N95 information was calculated from the signals. In the third stage, PERG derived features were integrated with visual acuity and demographic information. Finally, ML models and hybrid stacking ensemble were trained and evaluated under a patient level validation strategy.

### 2.1. Dataset

The data have been obtained from the open access dataset. The dataset, an electrophysiological signal collected at the Institute of Applied Ophthalmobiology (IOBA), University of Valladolid, Spain, between 2003 and 2022. PERG recordings were analyzed retrospectively [9,21]. The final dataset comprised 336 individual measurement sessions from 304 patients and 1354 eye trial level rows derived from these sessions. Each row corresponds to the eye and trial combination of a specific patient at a given visit. PERG recordings were obtained using a Metrovision MonPack 120 system in accordance with the ISCEV standards. A low-contrast black-and-white reversing checkerboard pattern at 1–2 Hz was used as the visual stimulus. Signals were recorded for 150 ms at a sampling frequency of 1700 Hz and yielding 255 samples per trial.

Demographic and clinical information for each record is provided in the file *participants_info.csv*. They also include visual acuity for the right eye (*VA-RE*) and the left eye (*VA-LE*) on the logMAR scale.

There are up to three structured diagnostic fields, Diagnosis1, Diagnosis2 and Diagnosis3. Visit-related flags are available, such as first versus follow-up visit and laterality. Treatment status is recorded. Narrative clinical comments are also included.

Demographic characteristics were calculated at the patient level, based on each individual's initial record. Accordingly, the mean age of the 304 patients included in the study at the first visit was 36.3 ± 18.5 years, and the median age was 37 years. The age range of the participants ranged from 4 to 86 years. The sex distribution was 51% female (155) and 49% male (149). From this perspective, the sample was sex balanced. An examination of the visit structure revealed that 281 patients (92.4%) had only a single visit, while 23 patients (7.6%) were followed up for multiple visits. Of these 23 patients, 19 had two separate PERG records, one had three, one had four, and two had five. In patients with multiple visits, the age difference between the earliest and latest records ranged from 0 to 13 years, with a median difference of approximately 2 years. Thus, age and date information appear consistent across repeated measurements of the same patient. The *rep_record* variable in the dataset represents the *ID_Record* numbers of previous visits for patients with follow-up, and these numbers match the actual records in the dataset.

Of the 336 clinical records, 106 were classified as normal and 230 as pathological. Conversely, of the 304 individuals participating in the study, 100 were classified as normal and 204 as pathological. Record level age, sex and logMAR visual acuity values were derived from the information reported in the *participants_info* table for each *ID_Record*. Demographic and visual acuity characteristics according to diagnostic groups in the dataset are given in Table 1.

Table 1. Demographic and visual acuity characteristics according to diagnostic groups.

| Group | Record | Participant | Age | Age Range | Female | Male | VA_RE (logMAR) | VA_LE (logMAR) |
|---|---|---|---|---|---|---|---|---|
| Normal | 106 | 100 | 33.6 ± 18.1 | 6 – 86 | 52 (49.1%) | 54 (50.9%) | 0.18 ± 0.47 | 0.19 ± 0.41 |
| Pathological | 230 | 204 | 38.7 ± 18.2 | 4 – 81 | 124 (53.9%) | 106 (46.1%) | 0.41 ± 0.57 | 0.38 ± 0.53 |
| Total | 336 | 304 | 37.1 ± 18.3 | 4 – 86 | 176 (52.4%) | 160 (47.6%) | 0.34 ± 0.55 | 0.32 ± 0.50 |

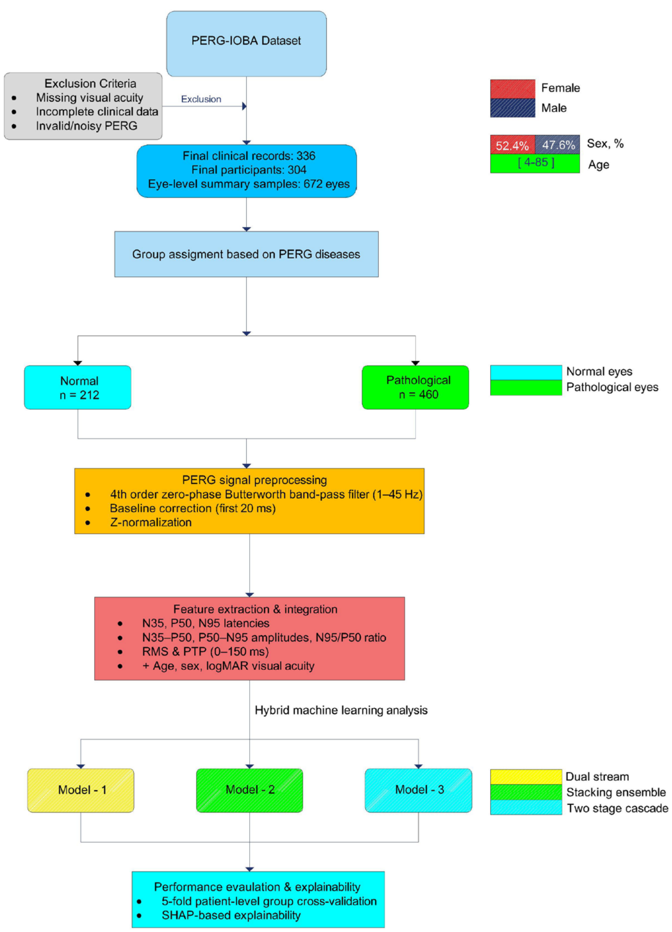


Figure 2. Flowchart of the study

The diagnosis1 field indicated the primary ophthalmologic diagnosis at the relevant visit, while the diagnosis2 and diagnosis3 fields indicated any accompanying comorbidities in the dataset. In this study, the primary diagnosis was the target variable in the main analyses, and the diagnosis1 field was used. Accordingly, records with diagnosis1 = Normal were grouped as normal, while records with all other primary diagnoses were grouped under the pathological group. Additional diagnoses in the diagnosis2 and diagnosis3 fields were retained as comorbidity information. However, to avoid distorting the primary diagnosis classification and creating heterogeneous subgroups, they were not included in the main group definition. The sample selection and group definition steps of the study are shown in Figure 2.

## 2.2. Preprocessing and feature engineering

Raw PERG signals may contain artifacts such as low-frequency baseline drift, electrical grid interference, high-frequency biological noise, and eye blinking. Therefore, all recordings were preprocessed according to the ISCEV standard before analysis.

In the first step, all signals were passed through a fourth-order, zero-phase Butterworth band-pass filter in the 1–45 Hz range. This band was used to reduce baseline drift below 1 Hz and high-frequency noise above 45 Hz [22]. The zero-phase approach prevented phase shifts that might have been caused by the delay structure of the N35, P50, and N95 components. Delay and amplitude measurements were made using filtered, baseline-corrected, and scaled signals in the 1–45 Hz range defined in the ISCEV standard. Raw signals were used for visual inspection only. For each visit and eye, denoised PERG waveforms were obtained, with the N35, P50, and N95 components reliably identified. This created a consistent feature set for both statistical analyses and ML models.

Baseline correction was performed by subtracting the average of the first 20 ms of each signal, moving the signal around zero after filtering [23]. This reduced DC shift and enabled a clearer assessment of the P50 and N95 components [24]. All signals were then scaled using z-normalization to reduce inter-individual amplitude differences and achieve a more balanced data structure for analysis. This step was performed to prevent amplitude-induced negative evaluation in subsequent ML models [25]. Three clinically important principal components were automatically identified after preprocessing. Time windows of 25–45 ms were used for N35, 45–70 ms for P50, and 80–130 ms for N95. Positive or negative extremums were selected within each window using derivative-based zero-crossing methods. Latency and amplitude values were calculated for each component. Furthermore, the amplitude differences between N35–P50 and P50–N95, as well as the ratio between these two amplitudes, were obtained as derived features. Additionally, RMS and peak-to-peak values were calculated for each signal in the 0–150 ms range. They were added to the dataset as time-domain features reflecting overall signal strength and waveform variability [25,26].

When multiple PERG trials were recorded from the same patient, at the same visit, and for the same eye (e.g., RE1–RE3, LE1–LE3), these signals were grouped at the (*Patient_ID*, *ID_Record*, *Eye*) level. A single summary value was created for each eye by taking the median of all latency, amplitude, and waveform summary parameters for the N35, P50 and N95 components. Thus, the final analysis dataset comprised a total of 672 eye-level samples. Each row represented a summary PERG measurement for a single eye at a given visit; 212 of these were classified as Normal and 460 as Pathological eyes. Derived features are summarized in Table 2. N35/P50/N95 latency and amplitude values and RMS were used because these are standard and interpretable PERG markers in clinical practice. The main goal of this study was to reliably implement patient-level validation. This approach reduces the number of independent training subjects. In this condition, deep learning with raw signals can more easily lead to overlearning. Therefore, this study focused on these features.

Table 2. PERG derived parameters and visual acuity of the corresponding eye by diagnostic group.

| Variable | Normal (n =212 eyes) | Pathologic (n =460 eyes) | Total (n= 672 eyes) |
|---|---|---|---|
| N35 latency (ms) | 30.6 ± 3.3 | 33.4 ± 3.8 | 32.5 ± 3.9 |
| P50 latency (ms) | 53.4 ± 4.0 | 54.9 ± 4.5 | 54.4 ± 4.4 |
| N95 latency (ms) | 103.8 ± 8.3 | 105.4 ± 9.6 | 104.9 ± 9.2 |
| N35–P50 amplitude, z | 2.05 ± 0.94 | 1.39 ± 1.11 | 1.60 ± 1.11 |
| P50–N95 amplitude, z | 3.29 ± 0.60 | 2.48 ± 1.18 | 2.74 ± 1.10 |
| N95/P50 amplitude ratio | 1.7 ± 3.0 | 3.8 ± 22.6 | 3.1 ± 18.8 |
| RMS 0–150 ms, z | 1.00 ± 0.00 | 1.00 ± 0.00 | 1.00 ± 0.00 |
| PTP 0–150 ms, z | 3.51 ± 0.25 | 3.49 ± 0.34 | 3.50 ± 0.32 |
| VA (corresponding eye, logMAR) | 0.19 ± 0.44 | 0.39 ± 0.55 | 0.33 ± 0.52 |

### 2.3. Machine learning models

The aim was to classify normal and pathological eyes using PERG derived features and clinical variables. In all models, the input was summary data rows generated for each eye. Raw PERG waveforms were not used in the ML phase. Only electrophysiological properties such as latency, amplitude, ratio, root mean square (RMS) and peak to peak (PTP), as well as demographic and visual acuity information obtained from signals filtered according to the ISCEV standard, were used. For comparison, LR, RF, gradient boosting tree-based methods, and so on, were also tested. However, the focus of this study is to examine the performance of three different hybrid models.

Model 1 processes PERG derived features and clinical/visual acuity features in two separate MLP streams and combines the representation vectors obtained from these two streams to a single output layer. Model 2 is a stacked ensemble structure built on the probability outputs of the base classifiers, which are RF, XGBoost, LightGBM, and a small MLP. The probabilities generated by the base models are combined with a LR model. Model 3 is a two-stage cascade structure: in the first stage, a more simplified model performs high-precision screening, and cases in which the model is highly confident are directly classified; uncertain (gray area cases) are directed to the advanced hybrid model in the second stage. In all models, the output is the probability that the relevant eye is in the Pathological class. The basic characteristics of Models 1–3 are summarized in Table 3.

Table 3. ML specifications.

| Model | Name | Input type | Key components | Main idea |
|---|---|---|---|---|
| Model 1 | Dual-stream feature fusion | PERG derived features + clinical/VA | Two separate MLPs (PERG and clinical branches), feature fusion layer, logistic output layer | To process PERG and clinical information in two separate streams and merge them into a shared representation |
| Model 2 | Stacking-based hybrid ensemble | Combined feature vector (PERG + clinical) | Base models: RF, XGBoost, LightGBM, small MLP; meta-model: logistic regression | To combine the predictions of different algorithms through a higher-level meta-model |
| Model 3 | Two-stage cascade classifier | Combined feature vector | Stage 1: simple model (e.g. XGBoost); Stage 2: Model 1 or Model 2 | To perform initial screening in the first stage and route challenging gray-zone cases to a more complex model in the second stage |

The data row obtained for each eye consisted of the following two subgroups. And actual class labels were defined for each eye.

- $X_{PERG} \in \mathbb{R}^{d_p}$ - PERG derived features
- $X_{clinic} \in \mathbb{R}^{d_c}$ - Clinical and demographic features
- $y \in \{0,1\}$ - $y = 0 \rightarrow Normal,$
  $y = 1 \rightarrow Pathologic$

Each model produces a predicted probability from these inputs.

$$\hat{p} = P(y = 1 | X_{PERG}, X_{clinic}) \tag{1}$$

And binary decision;

$$\hat{y} = \begin{cases} 1, & \hat{p} \geq \tau \\ 0, & \hat{p} < \tau \end{cases} \tag{2}$$

### 2.3.1. Model 1 – Dual stream feature fusion

PERG features and clinical/visual acuity are processed simultaneously, but on two separate streams. The representation vectors obtained from these two streams are then concatenated and fed into a single classification layer. Therefore, the input streams are PERG ($z_{PERG}$) and clinical ($z_{clinic}$) stream. Here, PERG consists of seven numerical features including N35, P50, and N95 latencies; N35–P50, P50–N95 amplitudes; N95/P50 ratio, and 0–150 ms PTP value (excluding the RMS value because it remains constant across eyes). Clinical stream includes the patient's age, logMAR visual acuity (VA) of the corresponding eye and sex.

The PERG stream was processed by a MLP with two hidden layers of 32 and 16 neurons, while the clinical stream was processed by a separate single layer MLP with 16 neurons. The ReLU activation function was used in both streams. This process can be expressed as the transformation of raw inputs ($x$) into more abstract feature vectors ($z$) as follows:

- PERG flow

$$z_{PERG} = f_{PERG}(x_{PERG}; \theta_{PERG}) \in \mathbb{R}^{k_p} \tag{3}$$

- Clinic flow

$$z_{clinic} = f_{clinic}(x_{clinic}; \theta_{clinic}) \in \mathbb{R}^{k_c} \tag{4}$$

Here, $f_{PERG}$ and $f_{clinic}$ are MLPs containing 1–2 hidden layers with ReLU activation. $\theta_{PERG}$ and $\theta_{clinic}$ show the parameters of the respective networks. The representation vectors from the two streams were then combined.

$$\boldsymbol{h} = [z_{PERG}; z_{clinic}] \in \mathbb{R}^{k_p+k_c} \tag{5}$$

The combined vector (**h**) is passed through a dense layer of 32 neurons and the pathology probability ( $\hat{p}$) is calculated with the final sigmoid output neuron

$$\hat{p} = \sigma(\boldsymbol{w}^T h + b) \tag{6}$$

Here $\sigma(\cdot)$ is the logistic sigmoid function. $\boldsymbol{w}$ and $b$ represent the weight and bias terms of the combined layer.

### 2.3.2. Model 2 – Stacking ensemble

In this model, multiple base classifiers are trained on the same inputs. The resulting output probabilities are combined by a higher-level meta-model, thus combining the strengths of different algorithms into a single hybrid structure. Let's use a total of $K$ base classifiers in this model.

$$f_1, f_2, f_3, \ldots f_K \tag{7}$$

Each of the combined input vectors

$$\mathbf{x} = [x_{PERG}; x_{clinic}] \tag{8}$$

In the base layer (Level-0), the system consists of four different base classifiers trained on the combined input vector ($x$). These classifiers are RF, XGBoost, LightGBM and the MLP Classifier. RF is used for its explainability and resistance to overfitting. XGBoost is preferred for its gradient boosting ability to model complex boundaries. LightGBM is used as a neural network component for its success with imbalanced data and to capture nonlinear relationships in MLP. During the training phase, these probabilities are generated using cross validation to prevent data leakage. Generates an estimated probability based on it;

$$p_k = f_k(x), \qquad k = 1,2,\ldots,K \tag{9}$$

$$\mathbf{p} = [p_1, p_2, \ldots, p_k]^T \in \mathbb{R}^K \tag{10}$$

In the meta layer (Level-1), the prediction probabilities generated by the base learners are fed into the LR meta-model (L2-penalized, class-weighted) as a meta-feature matrix. The meta-model combined these inputs to produce the final calibrated probability.

$$\hat{p} = \sigma(\beta_0 + \beta_1 p_1 + \beta_2 p_2 + \cdots + \beta_K p_K) \tag{11}$$

Here, $\beta_0$ is the constant term and $\beta_K$ is the coefficients of the meta model. If the meta model is a small MLP, this expression can be represented by the $g(\mathbf{p}; \theta_g)$ more general function.

$$\hat{p} = g(\mathbf{p}; \theta_g) \tag{12}$$

During the training phase, the meta-model was trained using the out-of-fold predictions of the base classifiers obtained through cross validation. This aimed to enable the meta-model to learn the error patterns of each base model and compensate for these errors.

### 2.3.3. Model 3 – Two-layer cascade classifier

In this model, a high-sensitivity initial screening is performed using a simpler and faster classifier. Eyes that the model is highly confident in are classified directly, while eyes in the "gray area" border are sent to the more complex model in the second stage. This establishes a structure similar to the screening-detailed examination logic used in clinical practice. First stage classifier;

$$p_1 = f_1(x) \tag{13}$$

Here, $f_1$ is usually a simpler model. For example, LR or a single XGBoost model. Two thresholds are defined:

$\tau_{low}$ : Safe limit in favor of normal
$\tau_{high}$ : Safe limit in favor of pathological

Decision rule:

$$\hat{y} = \begin{cases} 0, & p_1 \leq \tau_{low} \\ 1, & p_1 \geq \tau_{high} \\ uncertain, & \tau_{low} < p_1 < \tau_{high} \end{cases} \tag{14}$$

In this rule, $p_1 \leq \tau_{low}$ is considered normal and $p_1 \geq \tau_{high}$ is considered pathologic. For cases that remain uncertain, the second stage is activated. A richer model is chosen for the second stage. For example, the dual-stream network of Model 1 or the stacking structure of Model 2.

$$p_2 = f_2(x) \tag{15}$$

Final decision:

$$\hat{y} = \begin{cases} 0, & p_1 \leq \tau_{low} \\ 1, & p_1 \geq \tau_{high} \\ 1, & \tau_{low} < p_1 < \tau_{high} \text{ and } p_2 \geq 0.5 \\ 0, & \tau_{low} < p_1 < \tau_{high} \text{ and } p_2 < 0.5 \end{cases} \tag{16}$$

Thanks to this structure, the first stage functions more as patient screening/elimination, while the second stage serves as a detailed evaluation of difficult cases.

### 2.4. Model training, cross validation and hyperparameter optimization

In this study, all models were trained using the same data splitting strategy and similar training steps. The aim was to fairly compare models in the classification of normal and pathological eyes and to prevent patient level data leakage.

In the PERG-IOBA dataset, some patients had multiple clinic visits and multiple eye records. Therefore, training and testing were separated at the patient level. All eyes and visits from the same patient were always kept in the same cluster (training only or testing only). This prevented records from the same individual from being distributed across both training and testing sets, allowing model performance to be evaluated more closely to clinical reality. No patients were present in both the training and test sets. The risk of identity learning, where models memorize subject-specific features rather than generalize to pathology was eliminated.

Data splitting used a method similar to the *StratifiedGroupKFold* approach. The patient ID (*Patient_ID*) was used as the group variable, and the eye-level diagnostic label (Normal/Pathological) was used as the class variable. In the outer loop, the entire dataset was split into five folds at the patient level. In each outer layer, four folds were used for training, and one fold was used for independent validation (testing). Thus, each eye was tested at least once in a layer that had never been "seen" by the model, and the reported performance metrics were calculated as the average of these five outer folds.

PERG derived features (*N35_ms, P50_ms, N95_ms, Amp_N35P50_z, Amp_P50N95_z, Ratio_N95P50, PTP_0_150_z*) and clinical characteristics (*age, sex, logMAR visual acuity of the corresponding eye, etc.*) were combined into a single combined feature vector. In models using LR and MLPs, standard scaling z-normalization was applied to continuous variables. No additional scaling was performed for RF, XGBoost and LightGBM tree-based methods. The sex variable was coded using the One-Hot Encoding method.

All scaling and transformations were defined using the scikit-learn pipeline. Importantly, the standardization parameters, mean and variance, were calculated only in the training folds and applied to the validation folds to strictly prevent data leakage. Because class imbalance was moderate for 212 normal eyes and 460 pathological eyes, and no resampling was performed. Instead, the contribution of the minority class was increased using the class_weight="balanced" option in the LR and tree-based models. This prevented the models from being biased towards predicting only the dominant class. Hyperparameter tuning for each model was performed using inner cross validation within the outer 5-fold structure. The base structure and meta-classifier hyperparameters were tuned using the grid search method. Balanced accuracy and AUC were used as optimization criteria. This approach ensured that model selection remained completely independent of the patient-level, decoupled test set. In each outer layer:

- The four-fold training data was split into three-fold inner cross validation, also at the patient level.
- RandomizedSearch or limited grid search (GridSearch) was applied over predefined hyperparameter ranges.
- Balanced accuracy and ROC-AUC values were evaluated together in the inner cross validation; balanced accuracy was the main selection criterion.
- For each model, the hyperparameter combination yielding the highest average score in the inner cross validation was selected, and this tuning was applied to the model fully retrained on the outer layer's training data.

Finally, this model was tested on the outer layer's validation data, and the resulting performance metrics were recorded. The hyperparameter ranges used and the best values selected are summarized in Table 4.

- In general, reasonable ranges consistent with the literature were selected for:
- Regularization coefficient (C) and penalty type for LR;
- Number of trees, maximum depth, and minimum number of samples per leaf for RF;
- Learning rate, maximum depth, number of trees, and subsampling ratios for XGBoost and LightGBM;
- Number of hidden layers, number of neurons, learning rate, and dropout rate for MLPs.

### 2.5 Performance metrics

In this study, classification was considered as a two-class problem at the eye level. These are Normal and Pathological eyes. Because there was a moderate imbalance between the classes (212 Normal, 460 Pathological eyes), balanced accuracy was used as the main evaluation metric. Balanced accuracy was defined as the average of the sensitivity and specificity values. In each outer layer, a confusion matrix was obtained by comparing the model's predicted classes with the actual labels. Here:

- *TP (true positive):* Correct classification of the Pathological eye as Pathological
- *FP (false positive):* Incorrect assignment of the Normal eye to the Pathological class
- *TN (true negative):* Correct classification of the Normal eye as Normal
- *FN (false negative):* Incorrect assignment of the Pathological eye to the Normal class

Table 4. Hyperparameter search ranges.

| Model | Hyperparameter | Description | Search range / candidate values | Best value |
|---|---|---|---|---|
| Logistic Regression | C | Inverse regularization strength | 0.01 – 10 (log-scale, continuous) | 1.57 |
| | penalty | Penalty type | {L2} | L2 |
| Random Forest | n_estimators | Number of trees | 200 – 1000 (integer) | 356 |
| | max_depth | Maximum tree depth | {None, 5, 10, 20} | 20 |
| | min_samples_leaf | Minimum samples per leaf | {1, 2, 3, 4} | 1 |
| | max_features | Proportion of features considered at each split | {sqrt, log2} | sqrt |
| XGBoost | n_estimators | Number of trees | 100 – 800 | 400 |
| | max_depth | Maximum tree depth | 2 – 6 | 4 |
| | learning_rate | Learning rate | 0.01 – 0.30 | 0.10 |
| | subsample | Row subsampling ratio | 0.6 – 1.0 | 0.8 |
| | colsample_bytree | Column subsampling ratio | 0.6 – 1.0 | 0.8 |
| LightGBM | n_estimators | Number of trees | 100 – 800 | 350 |
| | num_leaves | Number of leaves | 15 – 63 | 31 |
| | learning_rate | Learning rate | 0.01 – 0.20 | 0.05 |
| | feature_fraction | Feature subsampling ratio | 0.6 – 1.0 | 0.8 |
| | bagging_fraction | Row subsampling ratio | 0.6 – 1.0 | 0.8 |
| MLP (tabular) | hidden_layer_sizes | Hidden layer / neuron configuration | {(64), (64, 32), (128, 64)} | (64, 32) |
| | activation | Activation function | {ReLU, tanh} | ReLU |
| | alpha | L2 penalty coefficient | 1e-5 – 1e-2 | 1e-4 |

Using these terms, the basic metrics were calculated. Sensitivity represents the proportion of pathological eyes the model correctly identifies and corresponds to the recall criterion for the Pathological class. Specificity represents the proportion of normal eyes correctly recognized as normal and can be thought of as the recall criterion for the Normal class. Accuracy reflects the ratio of correctly classified examples to the total number of examples when all eyes are considered. To compensate for imbalance between classes, balanced accuracy, used as the primary criterion in this study, is interpreted as the simple average of sensitivity and specificity, ensuring that both classes are represented with equal weight. Precision describes the proportion of eyes labeled pathological by the model that are actually pathological and is particularly important for assessing the clinical impact of false positives. The F1 score combines sensitivity and precision into a single value, providing a balanced combination of these two metrics; it is particularly useful for summarizing the model's overall performance in unbalanced class distributions. In these definitions, positive classes are considered pathological eyes unless otherwise specified.

In addition to these threshold-dependent metrics, the area under the ROC curve (ROC-AUC) of the models was also calculated. ROC-AUC was obtained using the continuous probability scores (probability of belonging to the pathological class) generated by the model. For Models 1 and 2, this probability corresponds to the output of the sigmoid function in the output layer. In Model 3, which has a two-stage cascade structure, the effective probability was defined by considering the thresholds in the first stage and, if necessary, the output of the second-stage model (details are explained in the relevant method section). For each model, these metrics were calculated separately on each of the five outer layers, and the final results were reported as the mean (and standard deviation, where appropriate) of the values across these five test layers.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

$$Balanced\ accuracy = \frac{Sensitivity + Specificity}{2} \tag{18}$$

$$Precision = \frac{TP}{TP + FP} \tag{19}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{20}$$

$$Specificity = \frac{TN}{TN + FP} \tag{21}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{22}$$

### 2.6. Explainability and statistical analysis

In this study, explainability analyses were conducted to examine not only the accuracy of the ML models but also the features they used to make their decisions. Furthermore, classical statistical tests were used to evaluate the differences between the Normal and Pathological groups and to statistically test the performance differences between the models.

Explainability analyses were first conducted on the best-performing hybrid model. For this purpose, the SHAP approach was used to examine the relationship between the probability estimates generated by the model for each eye and the input features. SHAP coefficients indicate the direction and magnitude of each feature's contribution to the probability that an individual eye is in the Pathological class.

First, all eyes were considered together, and the average of the absolute SHAP values for each feature was calculated, and a global importance ranking was derived using these values. This ranking compared the contribution of PERG derived parameters (such as N35, P50, and N95 latencies, amplitude differences, N95/P50 ratio, PTP, etc.) and clinical variables (such as age, sex, logMAR visual acuity, etc.) to model decision-making. Then, individual SHAP profiles were examined for a select few sample eyes (correctly classified typical normal and pathological cases, and incorrectly classified borderline cases) to interpret how the model "thinks" clinically based on these examples.

Classical feature importance levels (importance based on within-tree splits and permutation-based importance) were also calculated for the tree-based baseline models (RF, XGBoost, LightGBM). These results were evaluated together with the findings from the SHAP analysis, and the relative weights given to PERG and clinical features by different model families were compared. The resulting explainability results are presented with relevant figures and tables in the results section.

Differences between normal and pathological eyes for demographic, clinical, and PERG derived variables were assessed using appropriate parametric or nonparametric tests, depending on the distribution of the variable. The distribution of continuous variables was checked using the Shapiro–Wilk test and visual inspection (histogram, Q–Q plots). For variables approaching a normal distribution and with similar variances, the Student t-test was used for between-group comparisons, while for variables that did not meet distribution assumptions, the Mann–Whitney U test was used. Results were summarized as mean ± standard deviation or median (interquartile range), as appropriate. Group comparisons for categorical variables were performed using the chi-square test or, in cases where expected cell numbers were low, Fisher's exact chi-square test.

To assess performance differences between ML models, the balanced accuracy, F1 score, and ROC-AUC values obtained by each model across the outer layers were compared. The DeLong test was used to compare ROC-AUC values to test the difference between curves obtained on the same test samples. This non-parametric method is appropriate for comparing correlated ROC curves because it accounts for the shared sampling variability when the same cases are evaluated by multiple models. For other metrics, paired statistical tests (e.g., paired t-tests or Wilcoxon signed-rank tests) were applied to pairwise model comparisons based on the values obtained in the five outermost layers. Where necessary, significance levels for multiple comparisons were checked using appropriate methods (e.g., Bonferroni or FDR correction).

All statistical analyses and visualizations were conducted in the Python environment (pandas, NumPy, SciPy, scikit-learn, statsmodels and SHAP packages). Two-sided p-values were used in all tests, and $p < 0.05$ was considered the statistical significance threshold.

## 3. Results

### 3.1. Participant and record level characteristics

A total of 672 eyes were included in the study. 212 were normal and 460 were eyes with hereditary retinal or macular pathology. Summary statistics for PERG derived waveform parameters and eye-level visual acuity by diagnostic group are presented in Table 2. Overall, N35, P50, and N95 latencies were prolonged. At the same time, the amplitudes of N35–P50 and P50–N95 were decreased in pathological eyes, suggesting a pattern consistent with impaired inner retinal function. Significant differences were observed between the diagnostic groups. Specifically, latencies (N35, P50, and N95) were prolonged in the pathological group compared to the normal group (e.g., N35: 30.6 ± 3.3 ms compared to 33.4 ± 3.8 ms). Simultaneously, N35-P50 and P50-N95 amplitudes were significantly reduced in the pathological eyes (P50-N95: 2.48 ± 1.18 compared to 3.29 ± 0.60), reflecting a pattern consistent with impaired inner retinal function. Visual acuity was also worse in the pathological group (higher logMAR values: 0.39 ± 0.55) than in the normal group (0.19 ± 0.44).

### 3.2. Machine learning models performance

In this section, the classification performances of three different hybrid ML architectures (dual-stream, stacking, and cascade) using PERG signal features and clinical variables were presented. A 5-fold group cross validation scheme, disaggregated at the patient level, was used to evaluate the generalization ability of the models. Diagnostic efficacy was evaluated using accuracy, balanced accuracy, sensitivity (for the pathological class), specificity (for the normal class), F1-score, and area under the ROC curve (AUC). These results are given at Table 5. Furthermore, feature importance levels and explainability outputs were examined to better understand the decision-making mechanisms of the models.

Model 1 employed a structure that processed PERG derived waveform features and clinical variables such as age, sex, and eye-level logMAR visual acuity in two separate hidden layers and combined them at the late stage. Typical DL studies report near-perfect accuracies, often due to within-patient data leakage. In this study, the patient level validation scheme provides a more clinically realistic performance estimate with approximately 71% accuracy. This confirms that the model is learning generalizable disease patterns rather than memorizing individual patient characteristics. The close sensitivity and specificity values indicated that Model 1 provided a relatively unbiased, balanced discrimination between normal and pathological eyes, providing a reference starting point for other models.

Table 5. Models' performance metrics.

| Model | Accuracy | Balanced accuracy | Sensitivity (Recall, pathological) | Specificity (Recall, normal) | F1-score | ROC-AUC |
|---|---|---|---|---|---|---|
| Model 1 | 0.695 ± 0.078 | 0.700 ± 0.081 | 0.689 ± 0.085 | 0.711 ± 0.110 | 0.754 ± 0.070 | 0.739 ± 0.088 |
| Model 2 | 0.714 ± 0.031 | 0.701 ± 0.056 | 0.735 ± 0.022 | 0.668 ± 0.120 | 0.779 ± 0.023 | 0.762 ± 0.080 |
| Model 3 | 0.717 ± 0.055 | 0.676 ± 0.076 | 0.797 ± 0.064 | 0.555 ± 0.146 | 0.793 ± 0.036 | 0.755 ± 0.075 |

The biggest risk encountered in DL models, especially with limited datasets, is overfitting. To verify the training stability of the developed Model 1 (dual-stream network), the learning curves during the 5-fold cross validation process were analyzed. As seen in the graph in Figure 3, the training and validation loss values exhibit a parallel decline. The fact that the difference between the two curves (the generalization gap) remains low throughout the training period and the validation error closely follows the training error demonstrates that the model does not memorize the data and has a high generalization ability. In figure 3, the solid blue line represents the mean training loss, while the dashed red line represents the mean validation loss. The shaded areas indicate the standard deviation (±SD). The close proximity of training and validation curves demonstrates the absence of significant overfitting.

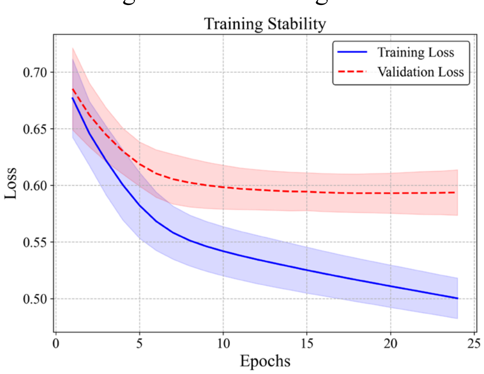


Figure 3. Learning curves of the Model 1.

Model 2 was based on combining the outputs of the base learners, consisting of RF, XGBoost, LightGBM, and a MLP with an LR-based meta-model. This hybrid architecture achieved the highest overall accuracy and highest ROC-AUC values (approximately 71% accuracy, AUC ≈ 0.76) over five-fold validation. The higher balanced accuracy value compared to Models 1 and 3 suggested that Model 2 was the architecture that minimized inter-class bias and most consistently distinguished normal from pathological eyes. Model 2 is a stacking ensemble and does not have an epoch-based learning curve like Model 1. Therefore, assessed its stability using fold-to-fold variability in the outer patient-level cross-validation. As reported in Table 5, Model 2 showed consistent performance across folds (e.g., accuracy 0.714 ± 0.031 and ROC-AUC 0.762 ± 0.080).

Model 3 was designed as a two-stage architecture simulating clinical triage logic. In the first stage, a fast-running XGBoost gate model directly classified normal and pathological cases with high confidence; Only gray-zone cases falling within the uncertain probability range, comprising approximately one-fifth of the total, were directed to the second-stage stacking model. This approach achieved the highest sensitivity (approximately 0.80) and the highest F1-score for the pathological class, making it the most robust model for capturing diseased eyes. Model 3 showed that the decrease in specificity generated more false positives, increasing the likelihood of patient loss. And demonstrated that it offered a screening-focused strategy.

Overall, Model 2 emerged as the best architecture in terms of accuracy and discriminative power (AUC). Model 3, with its high sensitivity and two-stage structure that reduces computational load. This offered a successful alternative, especially for pre-screening and cost-effective clinical triage scenarios. Model 1, which combines signal and clinical information in a dual-stream network, served as the fundamental reference model in these three architecture sets with its interpretable and stable performance. Figure 4 shows the ROC curves and Figure 5 shows the sensitivity-recall results.

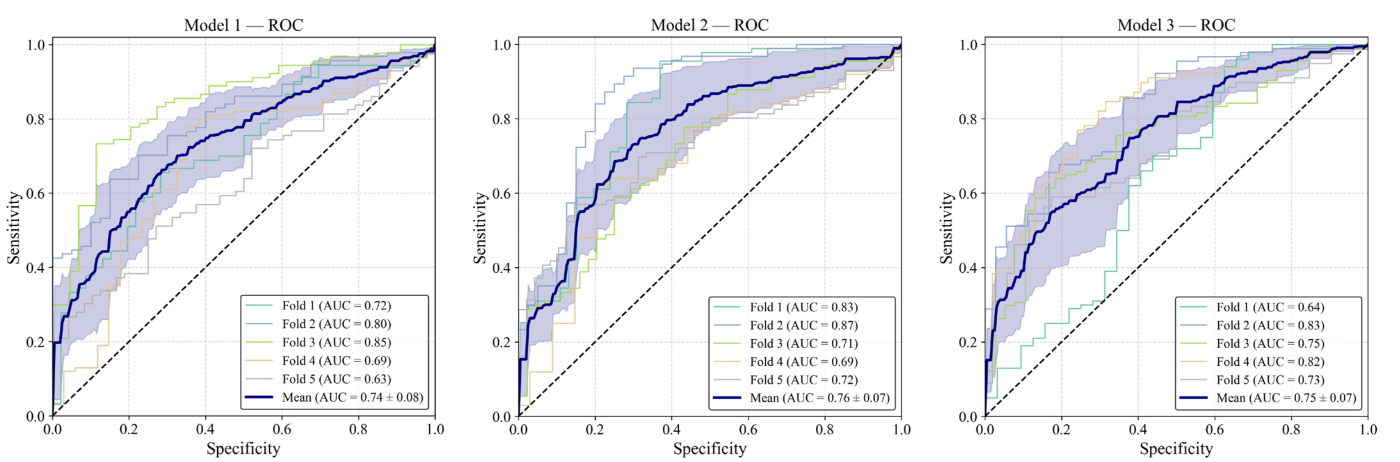


Figure 4. ROC curves of models

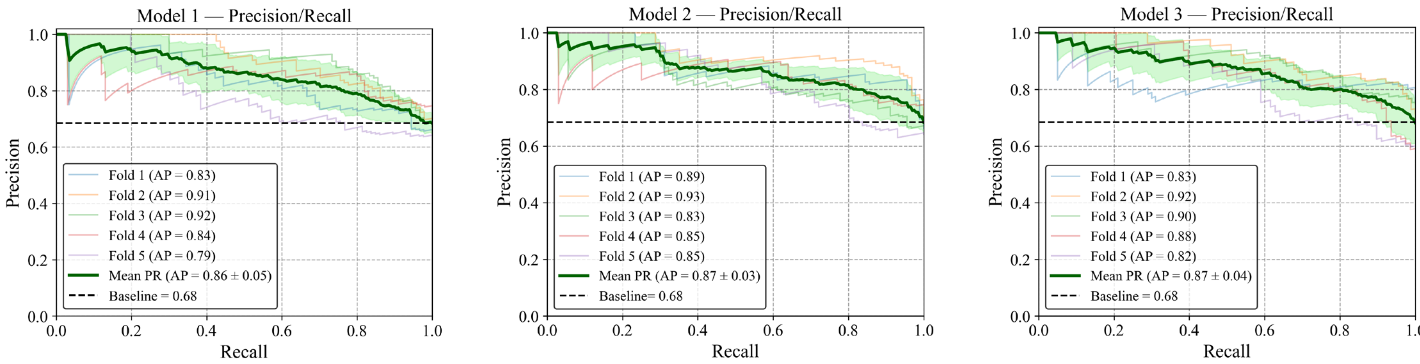


Figure 5. Precision recall results curves of models

In addition to discriminant performance (Accuracy/AUC), the reliability of the predicted probabilities was assessed using calibration curves. Ideally, a correctly calibrated model should align with a diagonal line indicating that the predicted probabilities match the observed disease frequencies. As shown in Figure 6, Model 2 exhibited the most stable calibration across five folds. Its global mean (red solid line) closely followed the diagonal line. This good calibration can be attributed to the LR meta-learning algorithm, which effectively converts the underlying model scores into realistic probability estimates. In contrast, Models 1 and 3 showed slightly higher variance across folds. This suggests small deviations in confidence levels across specific probability ranges.

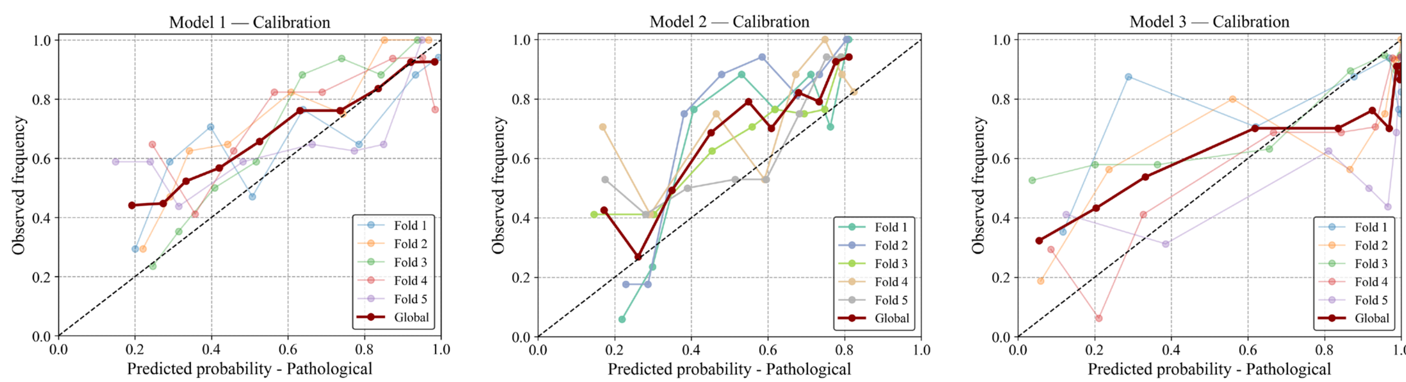


Figure 6. Calibration analysis of ML models

### 3.3. Feature importance and explainability

SHAP values were calculated to explain the decision-making process of the models. Figure 7 shows the overall feature importance and the effects of each feature on the model outputs. The superior biomarkers, P50-N95 amplitude (Amp_P50N95_z) and N35 latency (N35_ms), were identified as the most important features in all three models. The SHAP plots reveal that lower P50-N95 amplitude values (blue dots) are associated with a higher probability of pathology for a positive SHAP value. This is consistent with the clinical observation of amplitude reduction in retinal diseases. Conversely, higher N35 latency values (red dots) indicated that the model prediction should be evaluated as pathological class. In terms of model specific dependencies, Models 1 and 2 are a smooth mix of signal features (amplitude and latency) and clinical metadata. In particular, age emerged as the most important feature in Model 3.It was followed by signal parameters.

This demonstrates that the initial XGBoost or later stacking logic in a tiered architecture effectively uses age as the primary classifier for determining risk. It can be assumed that older patients are more frequently biased toward a pathological prediction to maximize sensitivity. For Visual Acuity, VA_eye_logMar contributed moderately to all models. Higher logMAR values (poorer vision) were consistently associated with an increased likelihood of disease.

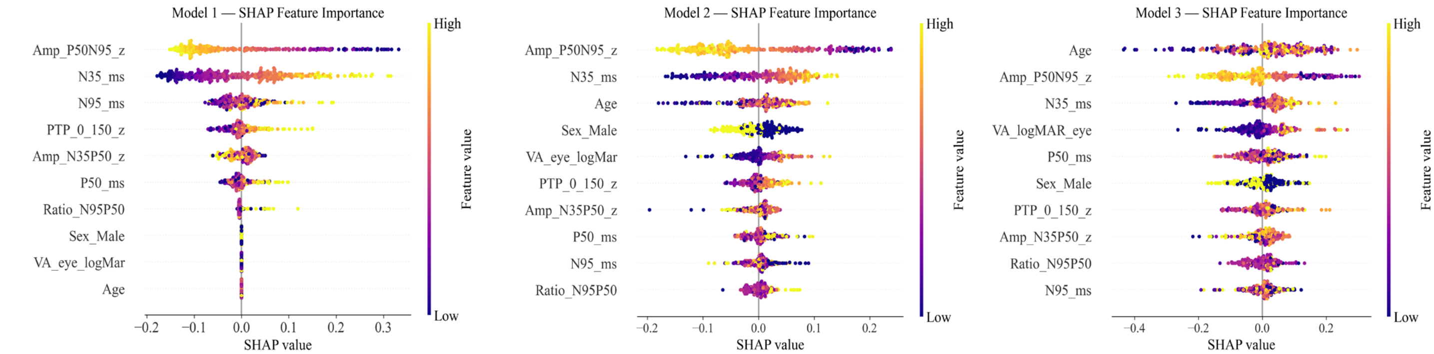


Figure 7. SHAP analysis

## 4. Discussion

It confirms that P50-N95 amplitude reduction is the most robust biomarker for retinal dysfunction, consistent with ISCEV standards. However, SHAP analysis revealed a critical finding: age is a dominant confounder. The model successfully learned to penalize amplitude expectations in older patients, mimicking an experienced clinician who knows that "normal" signals weaken with age. Direct comparisons with raw signal-based deep learning models were excluded from this study and will be addressed in a future study.

Unlike previous studies that reported >95% accuracy using random splitting (mixing left/right eyes of the same patient), our study accurately separated patients. The observed performance of 71–76% reflects the true difficulty of diagnosing retinal pathologies in a heterogeneous population with PERG alone. This demonstrates that PERG is a powerful adjunctive tool but should be part of a multimodal battery.

Model 2 offered the best stability, while Model 3 provided the best clinical utility. Capturing approximately 80% of pathologies, the Cascade model serves a potential "screening" role, filtering out obvious cases and flagging difficult cases for human review. An error analysis was conducted to understand the clinical distribution of cases misclassified by the model (false positives and false negatives). In the graph shown in Figure 8, the X-axis represents P50 latency, and the Y-axis represents P50-N95 amplitude. As expected, correctly classified pathological cases (green dots) are concentrated in the low amplitude and long latency region. However, the missed cases (false negatives), indicated by red crosses, appear to be mostly located in the gray zone where the normal and pathological boundaries intersect. This suggests that the model is very successful in capturing obvious pathologies but struggles with mild cases that fall within the physiological boundaries.

Model 3 had lower specificity, so it produced more false positives. This finding supports the necessity of the two-stage (cascade) structure in Model 3, as it is safer to refer patients in this gray zone to expert opinion rather than to automated decision-making. This may increase follow-up tests and workload. However, Model 3 was designed for triage. In triage, sensitivity is often prioritized to avoid missed cases. The decision threshold can be tuned based on clinical goals and available resources. False positives can be resolved with confirmatory evaluation.

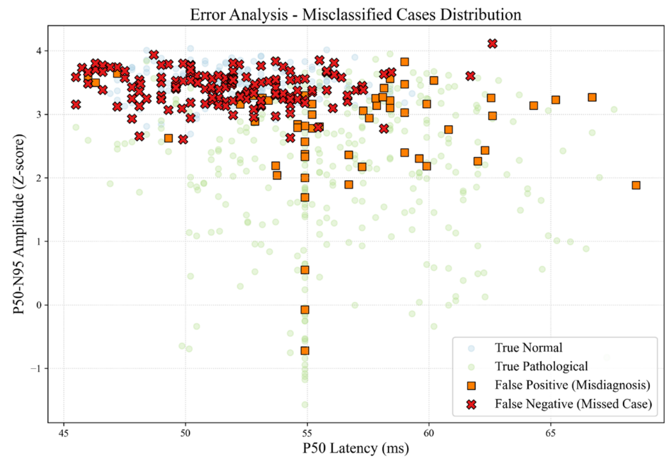


Figure 8. Error analysis scatter plot showing the distribution of predictions

### 4.1. Comparison with the literature

Studies on automatic detection of retinal dysfunction from PERG are growing. Many papers still rely on single-modality inputs and non-robust validation. Earlier work often compared N35/P50/N95 with classical statistics or simple cut-offs, which matches clinical practice but may miss nonlinear patterns in the data [3,10]. While this reflects clinical heuristics, it often fails to capture the complex, nonlinear interactions within high-dimensional data, as noted in recent physiological signal tracking studies [5,9,27].

Conversely, recent deep learning-based studies have reported classification accuracies exceeding 90% and, in some cases, reaching 98% [18,20]. In several of these reports, however, the split is performed at the eye or signal level rather than at the patient level. When both eyes (or repeated recordings) from the same subject appear in different folds, performance can be overestimated. For this reason, used strict patient-level separation. Under this setting, our accuracy (71–76%) reflects a more realistic estimate of generalization to unseen patients.

Furthermore, most AI models in the literature focus solely on signal morphology, ignoring clinical demographic metadata [28,29]. PERG amplitudes decrease with age, as noted in ISCEV-related clinical evidence [26]. By combining clinical variables with PERG features, models account for this variability. Consistently, SHAP highlighted age as a key predictor, which suggests that signal-only approaches may yield more false positives in older subjects if age effects are not modeled [16,30].

### 4.2. Methodological contribution

Unlike previous studies reporting very high accuracy values based on random data splitting, our study strictly applied patient-level separation. This approach prevents information leakage between training and test sets and provides a more realistic evaluation scenario. The obtained performance range of 71–76% reflects the actual difficulty of detecting retinal dysfunction using PERG signals in a heterogeneous clinical population. These findings emphasize that reliable AI-based diagnostic models require patient-level validation to avoid overly optimistic results.

### 4.3. Strengths and limitations

A major strength of this study is the proposed dual-stream architecture, which integrates PERG-derived signal features with basic clinical demographic data, resembling the way clinicians interpret electrophysiological findings in practice. In addition, the inclusion of an error analysis provides insight into model behavior in diagnostically uncertain cases.

This study also has some limitations. The analysis was conducted on a retrospective dataset with heterogeneous diagnostic categories, which may limit generalizability. Furthermore, although clinical metadata were incorporated, additional modalities such as retinal imaging (e.g., OCT) were not available and could further improve diagnostic performance. Future studies combining electrophysiological signals with imaging data in multi-center and prospective settings may enhance clinical applicability.

## 5. Conclusions

This study proposes a dual-stream ML framework for PERG-based retinal dysfunction detection with patient level validation. By combining features derived from PERG with baseline clinical data, the models were evaluated under conditions closer to clinical practice. Among the proposed approaches, the stacking ensemble model demonstrated the most stable performance, highlighting its suitability for supportive screening scenarios. Future studies aim to further improve diagnostic performance through the integration of complementary retinal imaging methods.

**Acknowledgment**

Not applicable.

**Nomenclature**

| | |
|---|---|
| *PERG* | Pattern Electroretinogram |
| *ERG* | Electroretinogram |
| *ISCEV* | International Society for Clinical Electrophysiology of Vision |
| *IOBA* | Institute of Applied Ophthalmobiology |
| *ML* | Machine Learning |
| *AI* | Artificial Intelligence |
| *XAI* | Explainable Artificial Intelligence |
| *SHAP* | SHapley Additive Explanations |
| *logMAR* | Logarithm of the Minimum Angle of Resolution |
| *VA* | Visual Acuity |
| *MLP* | Multilayer Perceptron |
| *RF* | Random Forest |
| *XGBoost* | eXtreme Gradient Boosting |
| *LightGBM* | Light Gradient Boosting Machine |
| *LR* | Logistic Regression |
| *RMS* | Root Mean Square |
| *PTP* | Peak-to-Peak |
| *ROC-AUC* | Receiver Operating Characteristic - Area Under the Curve |
| *N35, P50, N95* | Standard components and peaks of the PERG waveform |
| *Amp_P50N95_z* | Z-normalized P50-N95 amplitude |
| *N35_ms, P50_ms* | Signal latency values in milliseconds |
| $x_{PERG}$ | PERG-based feature vector |
| $x_{clinic}$ | Clinical and demographic feature vector |
| $y$ | Class label for pathological situation |
| $p$ | Predicted probability of pathology |
| $\tau$ | Classification decision threshold |
| $z_{PERG}, z_{clinic}$ | Representation vectors obtained for feature fusion |
| $\sigma$ | Logistic sigmoid function |

**Conflict of Interest Statement**

The author declares no competing interests.

**Ethics approval and consent to participate**

Data were collected during clinical activity and compiled for a specific project focused on the automated analysis of electrical signals obtained through ocular electrophysiology tests, approved by the IOBA research commission (approval 2021/47). All patients were informed about the potential use of their data for research purposes during ocular electrophysiological tests, ensuring compliance with general data protection regulations for informed consent.

**Consent for publication**

Not applicable.

**Relevant guidelines and regulations**

This study was conducted according to the ethical standards of the institutional research committee and the 1964 Helsinki Declaration and its later amendments.

**Availability of data and materials**

All data analyzed in this study are publicly available. The PERG-IOBA dataset can be accessed via https://physionet.org/content/perg-ioba-dataset/1.0.0/ under the Creative Commons Attribution 4.0 International (CC BY 4.0) license. Fernández, I., Cuadrado Asensio, R., Larriba, Y., Rueda, C., & Coco-Martin, R. M. (2024). A Comprehensive Dataset of Pattern Electroretinograms for Ocular Electrophysiology Research: The PERG-IOBA Dataset (version 1.0.0). PhysioNet. RRID:SCR_007345. https://doi.org/10.13026/2t6a-xq52

Additional information generated or analyzed during the current study is included in this manuscript and its supplementary files.

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